ICT30016 IoT Track – Week 3

Getting Good with Arduino Programming

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The Arduino Programming Language

- "Arduino" is actually the name of the whole Electronics Development Ecosystem. It consists of:
  A. The Arduino Board consisting of a microcontroller running the Arduino Firmware
  B. The Arduino IDE (a free programming software to write scripts for the Firmware to run)
  C. The Arduino Language



- This week's focus is on helping you master the Arduino Language.
  - The primary syntax is based on C++ but is not exactly C++.
  - Has built-in libraries to operate common complex components.
  - You can download and access additional libraries for many other components.
  - A lot of non-Arduino microcontrollers have support for this language, but you must download and include the board type into the IDE first.

- Begin by familiarizing yourself with where to find help:
  1. The official Website - https://www.arduino.cc/ (find, download & Install the Arduino IDE)
  2. Reference for standard Arduino functions & libraries - https://www.arduino.cc/reference/en/
  3. Learn the language here - https://www.arduino.cc/en/Tutorial/HomePage

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Level 1: C Programming in Arduino (2 Hours)

Just like learning a new language, you must methodically read and try the examples in this order:

1. Program Structure of a Sketch (the Arduino source code file)
2. Data Types available
3. How to set Variables
4. Using Operators
5. Decisions using If..Else and Switch
6. Loops using for, while and do
7. Writing Functions
8. Using Strings and String Objects
9. Control Timing
10. Using Arrays



Caution!

Even if you are familiar with C/C++, you must go through these tutorials and try them out on Arduino IDE at least once.

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Level 2: Revisiting ESP32 Electronic Functions (1 hour)

Now that you are proficient in the C Syntax, review the tutorial examples in Week 2 to revise on how these functions work:

1. pinMode() – used in the setup() function to configure a pin to be INPUT or OUTPUT
2. digitalRead() – used to read an ON/OFF signal from a sensor.
3. digitalWrite() – used to send a ON/OFF signal to the actuator.
4. analogRead() – used to read a variable-voltage level signal from an Analog Pin.
5. analogWrite() – used to output a variable-voltage level signal to a PWM Pin.
6. Serial.begin() – used in the setup() function to start the Serial Communications feature.
7. Serial.print() – used to transmit a string through the COM line towards your PC.
8. Serial.println() – same as print(), but this one will end with a new line.

The full Reference can be found here: https://www.arduino.cc/reference/en/

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Level 3: Look for Examples

Depending on what you want to achieve in your ESP32 (or Arduino Uno), you can look for examples in primarily 2 places:

1. Built-in Examples
    - In Arduino Uno, go to File > Examples; browse and find an example file that can show you how to do a specific function or use a feature.
    - Explanation for each example can be found here: https://docs.arduino.cc/built-in-examples/

2. Tutorials Site and Example Uses
    - ESP32-related How Tos: https://randomnerdtutorials.com/getting-started-with-esp32/
    - Look for example videos like https://www.youtube.com/watch?v=tc3Qnf79Ny8

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Takeaway for this Week

1. Once and for all, it is time for you to master the Arduino Programming Language. Dedicate a day to go through C programming and Arduino functions.

2. Review how the ESP32 works and how it differs from Arduino Uno.

3. Remember, your equipment includes an Arduino Uno, if you want to implement non-Internet applications.

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